Problem Set 1

1) a) State space: Let us assume that we have a state i such that i ∈ {0,1,2,3,4,5,6}, where we can say that if i = 0, then no boxes were opened, and if i = 6, then all boxes are open

Initial State: i = 0

Actions: Open the next box at state i, if i < 6

Transition model: If we are able to open the next box, then that means that we can advance to the next state, that is i+1.

Goal test: i = 6

Cost: Each action costs 1 since at a given time, we are either opening the box or searching for the banana, and hence the total cost will be the number of boxes opened.

Solution: Open every box i, until i becomes 6 and when you reach the last box, open the box and pick the banana.

b) State space: It is a finite string that contains the alphabets {A, B, C, D, E}.

Initial State: We start from the input string ABABAECCEC

Goal: String must be exactly E.

Action: Figure out the positions that match the given set of rules and then replace them.

Transition Model: For every position that matches the rule, we replace that particular part of the string with a new string as per the rules in each step. For example, In our case, we know that AB=BC, so we can replace AB with BC wherever we find AB in our input string.

Cost: The cost is 1 unit for each rewritten string, since we first check whether the given position in the string applies to any of the rules that we have and then change it, if it does.

Solution: Traverse through the string and make the required adjustments at each step by looking at the rules.

c) State space: A grid of dimension nxn which has fixed pits and floor squares.

Initial State: No floor squares are painted, and we start from a specified position.

Goal: All squares are painted

Action: Paint or move to the adjacent unpainted square(Note: We cannot move into painted squares or pits).

Transition Model: You can either add paint to your square and leave your position unchanged or move to any adjacent square without making any changes to your square or position.

Cost: Since we can either paint the square or move to any adjacent unpainted square, the path cost is 1 unit, because we can just do 1 step at a time.

Solution: We move to every other square at least once and either paint it or move to the next adjacent unpainted square, until every square is colored.

d) State space: Over here, all the actions are taking place in a ship irrespective of the fact whether it's loaded or not.

Initial State: A ship containing all the containers, which are 13 containers in rows, 13 containers wide, and 5 containers tall.

Goal: To make the ship empty

Action: Pick a container from the top of a stack that is not empty and keep it on the dock

Transition Model: Each time we pick up a container from the stack the height of that particular stack decreases by 1.

Cost: Over here, the action cost will be 1 unit, since the crane is doing 1 step at a time, that is, first its moving to the location, then it is picking up the container and in the end its keeping it on the ground.

Solution: Over here, we are just traversing through all the containers in the ship as we remove our crane picking up 1 container at a time and placing it on the dock.

2a) 1,8,3,7,4,2,6,5

Numbers that are smaller than 1 (towards its right) = 0
Numbers that are smaller than 8 (towards its right) = 6 (8>3,8>7,8>4,8>5,8>6,8>2)
Numbers that are smaller than 3 (towards its right) = 1 (3>2)
Numbers that are smaller than 7 (towards its right) = 4 (7>4,7>2,7>6,7>5)
Numbers that are smaller than 4 (towards its right) = 1(4>2)
Numbers that are smaller than 6 (towards its right) = 1 (6>5)

Total: 13 inversions

b) i) S1: 1,4,2,7,3,6,8,5

Numbers that are smaller than 1 (towards its right) = 0
Numbers that are smaller than 4 (towards its right) = 2 (4>2,4>3)
Numbers that are smaller than 2 (towards its right) = 0
Numbers that are smaller than 7 (towards its right) = 3 (7>3,7>6,7>5)
Numbers that are smaller than 6 (towards its right) = 1 (6>5)
Numbers that are smaller than 8 (towards its right) = 1 (8>5)

Total: 7 inversions

Since the number of inversions in both the goal state and initial states are odd in this case, so we can say that this puzzle is solvable.

ii) 3,7,2,6,4,5,8

Numbers that are smaller than 3 (towards its right) = 2(3>2,3>1)
Numbers that are smaller than 7 (towards its right) = 5 (7>2,7>6,7>4,7>1,7>5)
Numbers that are smaller than 2 (towards its right) = 1(2>1)
Numbers that are smaller than 6 (towards its right) = 2(6>4,6>1,6>5)
Numbers that are smaller than 4 (towards its right) = 1(4>1)
Numbers that are smaller than 5 (towards its right) = 0
Numbers that are smaller than 8 (towards its right) = 0

Total inversions = 12 inversions

Since the number of inversions initial state is even and the goal state is odd in this case, we can say that this puzzle is not solvable.